

BEYOND THE PLAN

3 Pillars, 1 Practice

A Frontline Safety Framework
for Temporary Traffic Management

Industry Position Paper

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*Prepared for industry consultation
with Main Roads Western Australia and traffic management organisations*

1. Executive Summary

Temporary traffic management in Western Australia is supported by comprehensive standards — the AGTTM, MRWA Code of Practice, and AS 1742.3 provide detailed guidance for planning, designing, and implementing traffic management at work zones. These standards represent significant expertise and are fit for their purpose: describing how a job should be set up and how traffic should be managed.

Yet workplace incidents in traffic management continue to occur on jobs where the paperwork was complete, the crew was accredited, and the signs were on the ute. The gap is not in the standards. The gap is between what the plan describes at the start of a shift and what the crew experiences during it.

This paper proposes a frontline safety framework — Beyond the Plan: 3 Pillars, 1 Practice — that names and structures what experienced traffic controllers already know: that the three things you verify before you start (competent crew, sound plan, right equipment) are necessary but not sufficient. The fourth element — dynamic awareness, the continuous practice of reading conditions and being willing to act on what you see — is what separates a safe crew from a crew that had all the right paperwork but still got hurt.

The framework is grounded in evidence from MRWA Banner Alerts, mapped against existing legislative and regulatory requirements, and designed for practical adoption without adding compliance burden. It does not replace current standards. It provides the connecting tissue that makes them work in real time, on the road, when conditions change.

Key proposition: The three pillars are preconditions — without them, the job should not proceed. The practice is continuous — without it, the preconditions degrade as conditions change. Communication is the thread that holds all four together.

2. The Problem: When the Paperwork Is Perfect

2.1 The Gap Between Plan and Reality

A Traffic Management Plan is a snapshot. It describes the work zone as it was designed — the sign positions, the speed zones, the traffic control points. It assumes the weather stays the same, the traffic behaves predictably, and the hazards are known in advance. These assumptions hold at the moment the plan is written. They do not necessarily hold at 2:30 PM on a Tuesday when the wind picks up, the sun shifts, and the traffic volume doubles because a nearby school just finished for the day.

This is not a failure of the planning process. Plans cannot predict every condition. The purpose of a plan is to establish a safe baseline. The problem arises when the baseline is treated as the complete picture —

when the crew follows the plan right up to the moment someone gets hurt, and the investigation reveals that conditions had changed and nobody adjusted.

2.2 The Evidence: Banner Alert Case Studies

The MRWA Banner Alert system provides documented evidence of incidents that occurred on jobs where the standard preconditions were met. The following cases illustrate the gap:

Case 1: TC Slips on Soft Roadside Batter (EQ#57872)

A Traffic Controller exited their vehicle on the passenger side at 6:15 AM. Their feet slipped on a soft batter at the road shoulder. They grabbed the door and roll bar to right themselves, sustaining a shoulder injury that resulted in a Lost Time Injury.

Pre-start status: The crew was accredited. The TMP was in place. The signs were loaded and positioned. All three pillars were satisfied.

What the plan didn't account for: Low light at dawn combined with a soft roadside batter that looked stable but wasn't. This hazard only existed in that moment — it was not present when the site was assessed in daylight, and it would not be present after the ground dried. Dynamic awareness — scanning the ground surface before stepping out, especially in low light — would have identified the hazard before it caused injury.

Case 2: Worker Trips on Faded Wheel Stop (EQ#57539)

A worker tripped over a parking bay wheel stop bar that had faded over time, losing its visual contrast with the surrounding surface. The worker was not aware of the tripping hazard and sustained a leg injury.

Pre-start status: The wheel stop was there the entire time. The site had been in use for months.

What the plan didn't account for: The gradual degradation of a safety feature into a hazard. The wheel stop's colour markings had faded, reducing its visibility to the point where it became a trip hazard rather than a safety device. Nobody noticed because it blended in. Dynamic awareness — scanning for trip hazards on arrival and flagging them — would have caught this before someone was hurt.

Case 3: Sun Glare Changes Mid-Job (EQ#57762)

A traffic management job started with clear visibility. As the afternoon progressed, the sun shifted to a position that caused direct glare into the Traffic Controller's line of sight. The TC could no longer see oncoming traffic clearly, creating a significant safety risk.

Pre-start status: The TMP was correct for the conditions at the start of the shift. Signs were positioned properly.

What the plan didn't account for: The sun's position at 3:30 PM. The plan was written for the setup time, not the entire shift. Dynamic awareness — recognising that visibility had degraded and triggering a reassessment — would have led to repositioning the TC, adding a spotter, or pausing until the glare passed.

Case 4: Concrete Saw Strikes Worker (EQ#56820)

A quick-cut saw struck a worker in the face while cutting reinforced concrete pipe. The SWMS did not contain information on correct placement of the pipe to prevent pinching the blade. The prestart meeting form had been reused across multiple days instead of being completed fresh each day.

Pre-start status: A SWMS existed. A prestart form was completed. The crew had qualifications. On paper, the preconditions were met.

What the plan didn't account for: The SWMS was incomplete for the specific task. The prestart form was a formality rather than a genuine hazard assessment. This is a failure at both the organisational level (the system allowed a generic SWMS and reused prestart forms) and the crew level (nobody questioned whether the SWMS covered the work they were about to do). Dynamic awareness would have prompted the question: "Does our plan actually cover what we're doing right now?"

2.3 The Pattern

Across these incidents, a consistent pattern emerges:

- **Compliance:** The three preconditions were satisfied on paper
- **Change:** Conditions changed during the job, or a hazard existed that was not identified in the plan
- **Gap:** The crew did not adjust — either because they didn't notice the change, didn't feel authorised to act, or didn't have a process for reassessment

This pattern is not unique to Western Australia. It is consistent with safety research across high-risk industries, from construction to aviation to offshore operations. The pattern has a name in the safety literature: the gap between "work as imagined" (what the plan assumes) and "work as done" (what actually happens). The question is not whether this gap exists — it always does. The question is whether the crew has the awareness, the authority, and the process to bridge it.

3. The Framework: 3 Pillars, 1 Practice

The framework is built on a single observation: the things you verify before a job starts are different in kind from the thing you do while the job is happening. The three pillars are static — they are preconditions that, once verified, remain stable for the duration of the task. The practice is continuous — it begins when you arrive and does not stop until you pack down.

Pillar 1: Competent Crew

The question: Do we have people who know what they're doing?

Competence is not the same as accreditation. A Traffic Controller with a valid BWMT card has met the minimum training requirement, but competence on a specific job requires more: familiarity with the road, experience with the type of setup, and the judgement that comes from having worked in conditions similar to those expected on the day.

The distinction matters because accreditation is binary (you have the card or you don't) but competence is contextual. A TC who is highly competent on metropolitan signalised intersections may not be competent on a rural highway at night. A TC who has installed hundreds of lane closures may not be competent on a mobile convoy operation. Competence is always relative to the task, the location, and the conditions.

What this pillar requires:

- All crew members hold current accreditations appropriate to their role
- At least one crew member has experience with the specific type of setup required
- The crew has been briefed on the site-specific risks, not just the generic hazards
- Any crew member new to this type of work is paired with an experienced hand

It's broken when: The crew is accredited but nobody has worked this road before, or a TC is assigned to a setup type they've never done without supervision.

Pillar 2: Sound Plan

The question: Do we know what we're doing and how?

A sound plan is one that is specific to the site, current for the conditions, and understood by the crew. It is not a generic TGS pulled from a folder. It is not a TMP that was written for a different time of day, a different traffic volume, or a different road configuration. And it is not a document that sits on the dashboard unread while the crew sets up from memory.

The MRWA Code of Practice and AGTTM provide the design standards for traffic management plans. This pillar is not about the quality of those standards — it is about whether the plan that reaches the crew is fit for the specific job they are about to do. A plan that was correct when it was written but does not account for current conditions is not a sound plan for today.

What this pillar requires:

- The TMP and TGS are site-specific, not generic
- The plan accounts for the time of day, expected traffic volumes, and known environmental conditions
- The crew has read and understood the plan, not just received it
- The plan includes provisions for conditions that may change during the shift (weather, light, traffic)

It's broken when: The TMP is generic and doesn't match site conditions, the TGS was written for a different setup, or the crew hasn't actually read the plan.

Pillar 3: Right Equipment

The question: Do we have what the plan requires?

Having the right equipment means more than loading the signs listed on the TGS schedule. It means the signs are serviceable (not faded, damaged, or wrong-sized), the supplementary equipment is available (speed feedback signs, lighting towers, night wands), and there are sufficient reserves to adapt if conditions change during the shift.

The MRWA Code of Practice Section 6.16 mandates speed feedback signs at static worksites on Main Roads where workers on foot are not protected by barriers. Section 6.14 requires that all signs and devices are maintained in a serviceable condition throughout the work period. This pillar is about verifying compliance with these requirements before the job starts, not discovering their absence during it.

What this pillar requires:

- All signs and devices required by the TGS are loaded and in serviceable condition
- Supplementary equipment specified in the TMP is available (speed feedback signs, lighting, communication equipment)
- Equipment for contingencies is available (spare signs, additional delineation, wet weather gear)
- The crew has verified the equipment list against the plan, not just loaded from habit

It's broken when: Signs are loaded but the wrong size or faded beyond readability, the speed feedback sign that the TMP requires is back at the depot, or the crew loaded from memory without checking the schedule.

The Practice: Dynamic Awareness

“Dynamic Awareness is the practice of continuously reading the job while it's happening — and being willing to act on what you see.”

— Beyond the Plan Framework

Dynamic awareness is not a checklist. It is not something you tick off at the pre-start brief. It is the only element of the framework that is active for the entire duration of the job, from the moment the first crew member steps out of the vehicle to the moment the last sign is loaded back onto the ute.

The three pillars are static — once verified, they remain stable. Dynamic awareness is continuous. It exists precisely because the three pillars, no matter how well they are satisfied at the start, will degrade as conditions change. The wind picks up. The sun shifts. The traffic volume changes. A crew member

fatigues. A hazard that wasn't there yesterday appears. The plan doesn't cover it. That's where dynamic awareness becomes the safety net.

The Three Components of Dynamic Awareness

1. Reading Conditions

This is the observational foundation. Reading conditions means paying attention to what is actually happening, not what the forecast said this morning or what the TMP assumed. It includes monitoring weather changes (wind, rain, visibility), traffic behaviour (are drivers slowing down? is there congestion building?), road surface conditions (wet, soft, dusty), light conditions (sun glare, dusk, artificial lighting adequacy), and crew state (fatigue, distraction, morale).

Reading conditions is not passive observation. It is active scanning — deliberately looking for what has changed, not just what is there. The experienced TC who notices that the wind has strengthened enough to make a large sign unstable is reading conditions. The TC who walks the site on arrival and spots a pothole that wasn't there last week is reading conditions. The supervisor who sees a crew member struggling with fatigue at the two-hour mark is reading conditions.

2. Adjusting Controls

When conditions change, do you actually change what you're doing? Or do you push through because "the TMP says so"? Adjusting controls means having both the authority and the willingness to modify the traffic management setup in response to what you observe.

This is where the framework connects directly to existing standards. The MRWA TMP risk register section 3.2.1.2 states: "Traffic Management Personnel may adjust or change the site where a significant safety hazard is identified with appropriate approval and risk assessment." This sentence, buried in a risk register, is the regulatory basis for dynamic awareness. The framework gives it prominence and process.

Adjusting controls might mean repositioning a TC who can't see oncoming traffic due to sun glare. It might mean reducing the speed zone when drivers are ignoring the posted limit. It might mean adding a spotter when visibility drops. It might mean stopping the job entirely until conditions improve. The adjustment is always proportional to the observed change.

3. Stop Authority

The willingness to call a halt. Not after something goes wrong — when you feel something is off. This is where culture matters more than paperwork. If a TC feels they cannot call a stop without being reprimanded, the entire practice collapses. Stop authority is not a power — it is a permission that must be granted by the organisation, reinforced by supervisors, and exercised by every crew member without consequence.

The DuPont safety culture model, widely regarded as the gold standard in industrial safety, operates on a principle that any worker can stop any job at any time without question. This is not a sign of weakness or disruption — it is a sign of a mature safety culture. In traffic management, where conditions change

rapidly and the consequences of inaction are severe, stop authority is not optional. It is the mechanism that converts observation into action.

The Thread: Communication

Communication is not a fourth pillar or a fifth element. It is the thread that runs through all four components of the framework. The three pillars are verified through communication: pre-start briefings, equipment checks, plan reviews. Dynamic awareness is practised through communication: calling out what you see, radioing changes, speaking up about concerns, and — critically — calling a stop when something feels wrong.

Communication failures appear in banner alerts not as standalone causes but as contributing factors that amplify other failures. In EQ#56820, the prestart form was a communication failure — a form completed without genuine engagement. In EQ#57801, the chain recovery was a communication failure — nobody flagged that proper recovery equipment was not available. In each case, communication was the mechanism that could have bridged the gap between what was planned and what was actually happening.

For this reason, communication is not positioned as a separate pillar. Making it one would risk reducing it to a checkbox — "radio check, tick" — rather than recognising it as the continuous process that enables every other element of the framework.

4. Where This Lives in Existing Standards

The framework does not introduce new obligations. Every element maps to existing requirements in Australian and Western Australian standards. What the framework provides is structure — a way to organise existing obligations so that the gap between planning and practice becomes visible.

Framework Element	Existing Standard Reference	What the Framework Adds
Pillar 1: Competent Crew	AGTTM Part 7 — TC competency requirements MRWA CoP Section 8 — Accreditation WHS Act S27 — Officer due diligence	Distinguishes accreditation from competence. Makes context-specific experience visible, not just card status.
Pillar 2: Sound Plan	MRWA CoP Section 4 — TMP requirements AGTTM Part 3 — TGS design TMP 0922 Section 3.2.1 — Plan	Requires the plan to be verified as current and site-specific, not just present. Adds crew understanding

	adequacy	as a criterion.
Pillar 3: Right Equipment	MRWA CoP Section 6.16 — Speed feedback signs MRWA CoP Section 6.3 — PPE MRWA CoP Section 6.14 — Implementation and maintenance	Requires serviceability and sufficiency, not just presence. Adds contingency equipment.
Practice: Dynamic Awareness	TMP 0922 Section 3.2.1.2 — Site adjustment authority TMP 0922 Section 3.2.2 — Environmental conditions WHS Act S19 — Primary duty of care	Names and structures what is currently one sentence in a risk register. Provides the three-component model (read, adjust, stop).
Thread: Communication	MRWA CoP Section 6.14 — TTM operation AGTTM Part 7 — Communication protocols WHS Act S19 — Duty to consult	Positions communication as the enabler of all other elements, not a standalone compliance item.

The key mapping is between Dynamic Awareness and TMP Section 3.2.1.2, which states that Traffic Management Personnel may adjust or change the site where a significant safety hazard is identified. This single sentence is the regulatory authorisation for dynamic awareness. The framework transforms it from a buried provision into a visible, teachable, measurable practice.

5. The Maturity Model

Not all organisations practise the framework equally. The maturity model describes four levels of adoption, drawing on the Bradley Curve used in industrial safety culture assessment. It provides organisations with a way to assess where they currently sit and what the next step looks like.

Level	Name	Pillars Present	Practice Present	What This Looks Like
1	Compliance Only	2 and 3 (signs and plan)	No	The minimum. Signs are loaded, the TMP is on the dashboard. Crew competency is assumed based on accreditation. No process for reassessment during the job.
2	Competent	1, 2, and 3	Informal only	Qualified crew

	Individuals	(crew, plan, equipment)		doing the right thing individually. Experienced TCs may practise dynamic awareness personally, but it's not shared or supported by the organisation.
3	Coordinated Team	1, 2, and 3 (all verified)	Partially (no stop authority)	Crew communicates and coordinates effectively. Conditions are discussed. But stop authority is unclear or conditional — TCs hesitate to call a halt.
4	Adaptive Crew	1, 2, and 3 (all verified)	Yes (read, adjust, stop)	The full framework. The team reads conditions, adjusts controls when needed, and has clear stop authority. This is not dependent on individual experience — it is embedded in the crew's culture.

An organisation at Level 2 can pass an audit. An organisation at Level 4 is where nobody gets hurt. The difference is not in the paperwork — it is in the practice.

Most traffic management companies in Western Australia currently operate between Level 1 and Level 2. They have the signs and the plan. Some have competent crews. Very few have systematically addressed dynamic awareness as a safety practice, and even fewer have explicitly granted stop authority to their Traffic Controllers.

6. What This Means for Organisations

6.1 The Two Layers

The framework operates at two layers. The crew layer is what the Traffic Controller can control and practice: verifying the three pillars at the pre-start, practising dynamic awareness throughout the shift, and communicating continuously. The organisational layer is what the company must provide: the systems, resourcing, and culture that make the crew layer possible.

The banner alerts demonstrate what happens when the organisational layer fails. In EQ#56820, the prestart form was reused for a week because the company's system allowed it. In EQ#57801, proper recovery equipment was not available because the company did not equip the crew for contingencies. In each case, the crew could not practise what the framework requires because the organisation had not provided the conditions for it.

Layer	Responsibility	Examples
Crew	What the TC can control and practice	Verifying pillars at pre-start Reading conditions during the job Adjusting controls when conditions change Calling a stop when something feels wrong Communicating observations to the team
Organisational	What the company must provide	Systems that ensure plans are current and task-specific Resourcing that puts competent supervision on every job Culture that grants stop authority without consequence Learning loops that feed incidents back into practice Equipment that goes beyond the minimum schedule

6.2 Implementation Without Burden

The framework is designed to be adopted without adding compliance burden. It does not require new forms, new checklists, or new reporting obligations. It requires a shift in how existing processes are used:

- **Pre-start briefs:** The pre-start brief already happens. The framework asks that it covers all three pillars explicitly, not just the plan.

- **Site adjustments:** Site adjustments are already permitted under TMP 3.2.1.2. The framework asks that they are made proactively, not only after an incident.
- **Stop authority:** Stop authority is already implicit in the WHS duty of care. The framework asks that it is made explicit — every crew member knows they can call a stop, and every supervisor backs them up.
- **Incident learning:** Banner alerts are already issued by MRWA. The framework asks that they are used as learning tools, not just filed as compliance documents.

The shift is cultural, not procedural. The three pillars are already being checked in most organisations, though often implicitly. The practice of dynamic awareness is already being performed by experienced TCs, though often unconsciously. The framework makes both visible, teachable, and measurable — without requiring a new system to do it.

6.3 The Business Case

For organisations, the framework offers a clear business case beyond the moral imperative of keeping workers safe:

- **Reduced incidents:** Every Lost Time Injury costs the organisation in workers' compensation, lost productivity, investigation time, and replacement labour. The banner alerts document LTIs that occurred on jobs where the preconditions were met — meaning these were preventable with dynamic awareness, not with more paperwork.
- **Regulatory positioning:** Organisations that can demonstrate a mature safety culture — including the practice of dynamic awareness and explicit stop authority — are better positioned for MRWA pre-qualification, contract evaluation, and insurance assessment.
- **Crew retention:** TCs who feel empowered to speak up and adjust controls are more engaged, more likely to stay with the organisation, and more likely to develop into supervisors. The framework supports retention by valuing the judgement that experienced TCs already exercise.
- **Due diligence:** The WHS Act 2020 Section 27 requires officers to exercise due diligence, including acquiring and keeping up-to-date knowledge of hazards and risks. Dynamic awareness is the operational expression of that duty — it is how an organisation demonstrates that it is managing risks in real time, not just on paper.

7. The Evidence: Banner Alerts Mapped to the Framework

The following table maps MRWA Banner Alerts to the framework, showing which elements were satisfied and which were absent at the time of each incident.

Alert	Incident	Pillar 1	Pillar 2	Pillar 3	Practice	Root Gap
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		Crew	Plan	Equipment	Awareness	
EQ#57872	TC slipped on soft batter, 6:15 AM	✓	✓	✓	✗	Conditions changed (low light, soft ground). No scanning of ground surface.
EQ#57539	Worker tripped on faded wheel stop	✓	✓	✓	✗	Gradual hazard not identified. No arrival scan for trip hazards.
EQ#57762	Sun glare changed mid-job	✓	✓	✓	✗	Conditions changed (sun position). No reassessment triggered.
EQ#56820	Concrete saw strikes worker	✓*	✗	✓	✗	SWMS incomplete for task. Prestart form reused. Plan and practice both failed.
EQ#57801	Chain snaps through windscreen	✓	✗	✗	✗	Wrong recovery equipment. No one flagged the substitution. All three gaps.
EQ#57937	Truck rollover on wet surface	✓	✓	✓	✗	Conditions changed (wet road). Speed not adjusted. No reassessment.

** Crew was accredited but SWMS did not cover the specific task.*

The pattern is clear: the three pillars were satisfied in five of six incidents. The practice of dynamic awareness was absent in all six. The incidents were not caused by a lack of planning or equipment — they were caused by the gap between what was planned and what was happening, and the absence of a practice to bridge it.

This paper is presented for industry consultation. The framework is offered as a structure for a conversation that many in traffic management are already having informally: how do we keep people safe when the plan isn't enough? The 3 Pillars, 1 Practice framework does not claim to have all the answers. It claims to have named the right question.

Feedback is welcomed from Traffic Controllers, supervisors, safety managers, and regulatory bodies. The framework will be refined through industry input before any formal proposal is submitted to Main Roads Western Australia.